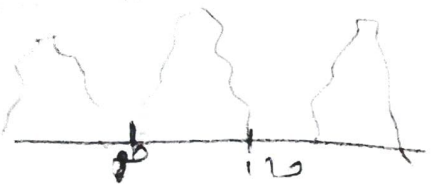
Q3) $FT\left(x_{(-T,T)}^{(t)}\right)$

$$= \int_{-\infty}^{\infty} x_{(-T,T)}^{(t)} \cdot e^{-j\omega t}\, dt = \int_{-T}^{T} e^{-j\omega t}\, dt$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$
$$-e^{i\theta} = -\cos\theta - i\sin\theta$$
$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{i}$$

$$= \frac{e^{-j\omega t}}{-j\omega}\Bigg|_{-T}^{T} = \frac{e^{-j\omega T} - e^{j\omega T}}{-j\omega} = \boxed{\frac{2\sin(\omega T)}{\omega}}$$ Answer

Q4) a)



Yunus Emre Fırtına 150117061

b) $\frac{1}{\pi}\sin(B_1 t)\cos(\omega_0 t) + \frac{1}{\pi}\sin(B_2 t)\cos(\omega_0 t)$

c) Convolution